AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 20 · STUDENT LESSON

# Conversation State and Memory

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

## Name the purpose. Track the correction. Clear on expiry.

Distinguish current application state from stale summaries and learned parameters.

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**Keep one habit:** Session policy → book-state trace → six retention cards → independent lab-session case.

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MAIN LESSON · SECTION 1

## What should the helper keep, replace or clear?

> Conversation state needs explicit scope, correction and reset rules.

### Your learning goals

I can explain the **mechanism** and terminology of conversation **state** and retention.

I can apply the **state**d procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

## Recall different meanings of memory

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Recall Model Memory vs Retrieval: an application's stored session fields are different from learned model parameters and retrieved source documents. Name the purpose and lifetime of a retained value.

## Your end goal

By the end, I can explain conversation **state** and memory using a traced example and a **state**d limitation.

### ▼ Earlier wording, alternatives and picture transcripts

#### Your learning target

#### Remember before reading

Without opening the old lesson, explain one key idea from User Need and AI Contribution. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Model Memory vs Retrieval. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Hallucination Failure Types. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 20: Conversation **State** and Memory

Products choose what context to keep, summarize, forget, or retrieve.



CREATOR MISSION

Design memory rules for a tutoring conversation.

Creator name and date:

**Write or draw your reasoning and evidence.**

**Readiness for the connected explanation**

**This week’s end goal**

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MAIN LESSON · SECTION 2

# Separate current fields, summaries and learned parameters

> Remembered wording is not automatically current or verified.

## Understand application state

Conversation state is an application record used across turns, such as a selected book or current topic. It is not evidence that a model permanently learned that preference. A summary is a derived copy and can become stale when a user corrects a value. Keep only fields needed for the task, attach their source turn, and define when they are replaced or cleared.

Product policies differ; this lesson uses the explicit toy session policy below.

Conversation **state** stores facts or prior interactions for later use. It needs scope, provenance, correction and **retention rule**s rather than indefinite accumulation.

## Use supplied fictional records

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use conversation **state** and memory to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

**A closer explanation**

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MAIN LESSON · SECTION 3

# Apply a visible state-transition contract

> Current value and audit history have different roles.

## Read the complete toy lifetime policy

Toy session policy: keep the one current approved selection only during this session. A correction replaces its active value; retain a minimal old→new audit event only until session end. A cached summary carrying the old selection must be refreshed or invalidated before use. On exit, clear the active selection, summary and session audit. A new session begins without a selection and asks for one.

Do not retain unrelated private text. This classroom contract does not claim to erase model training data or external logs in a real service.

## Connect room state and correction

This classroom assistant keeps an approved room ID only for the current session. A later correction replaces the active value but retains an audit event until session end.

Do not persist private text merely because it appeared in conversation.

## Respect scope and missing values

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the **state**d contract, record that condition explicitly instead of quietly replacing it with a convenient value.

## Explain the deciding check

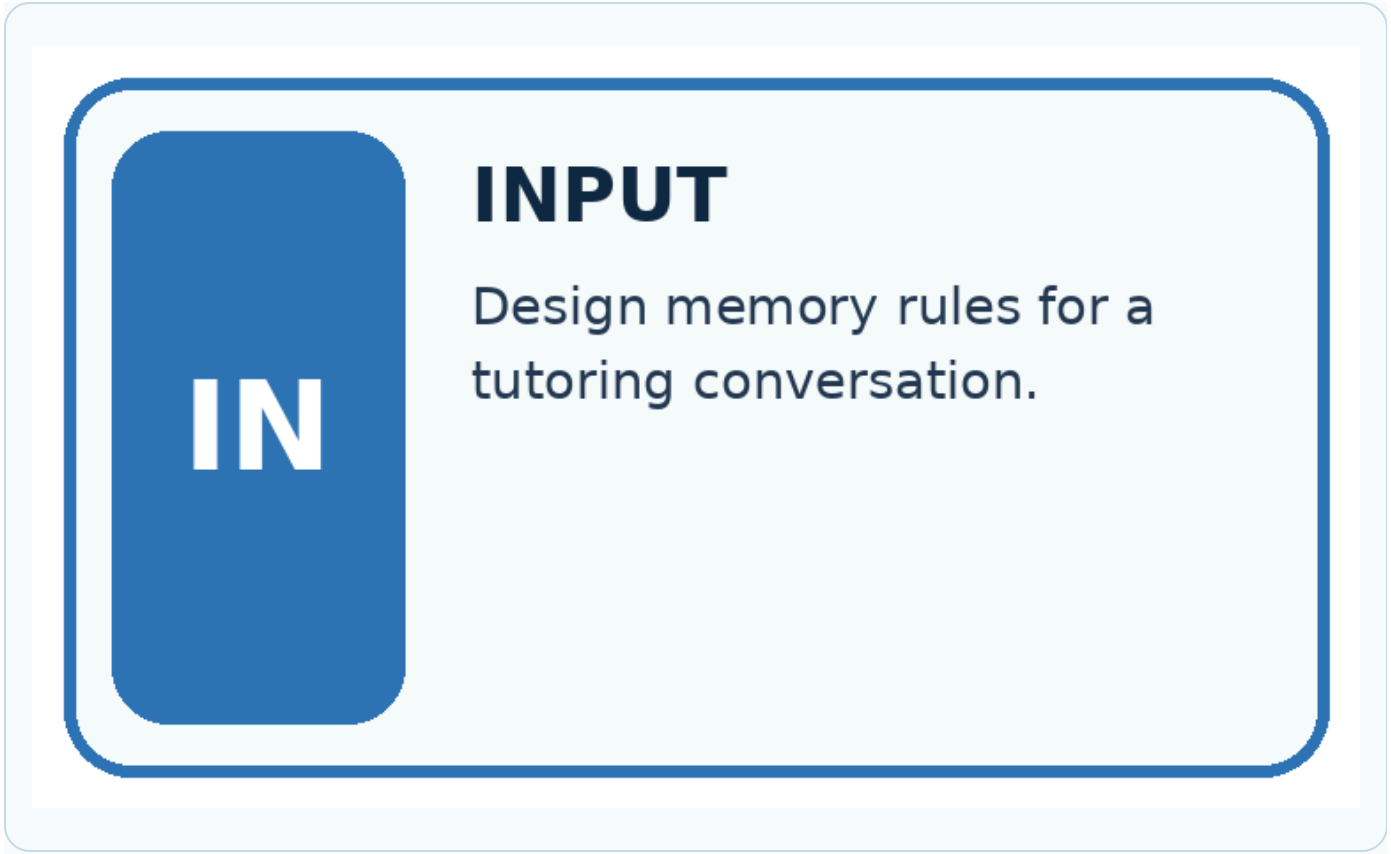
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

### ▼ Earlier wording, alternatives and picture transcripts

VISUAL **MECHANISM**

**Input** → **Process** → **Output**

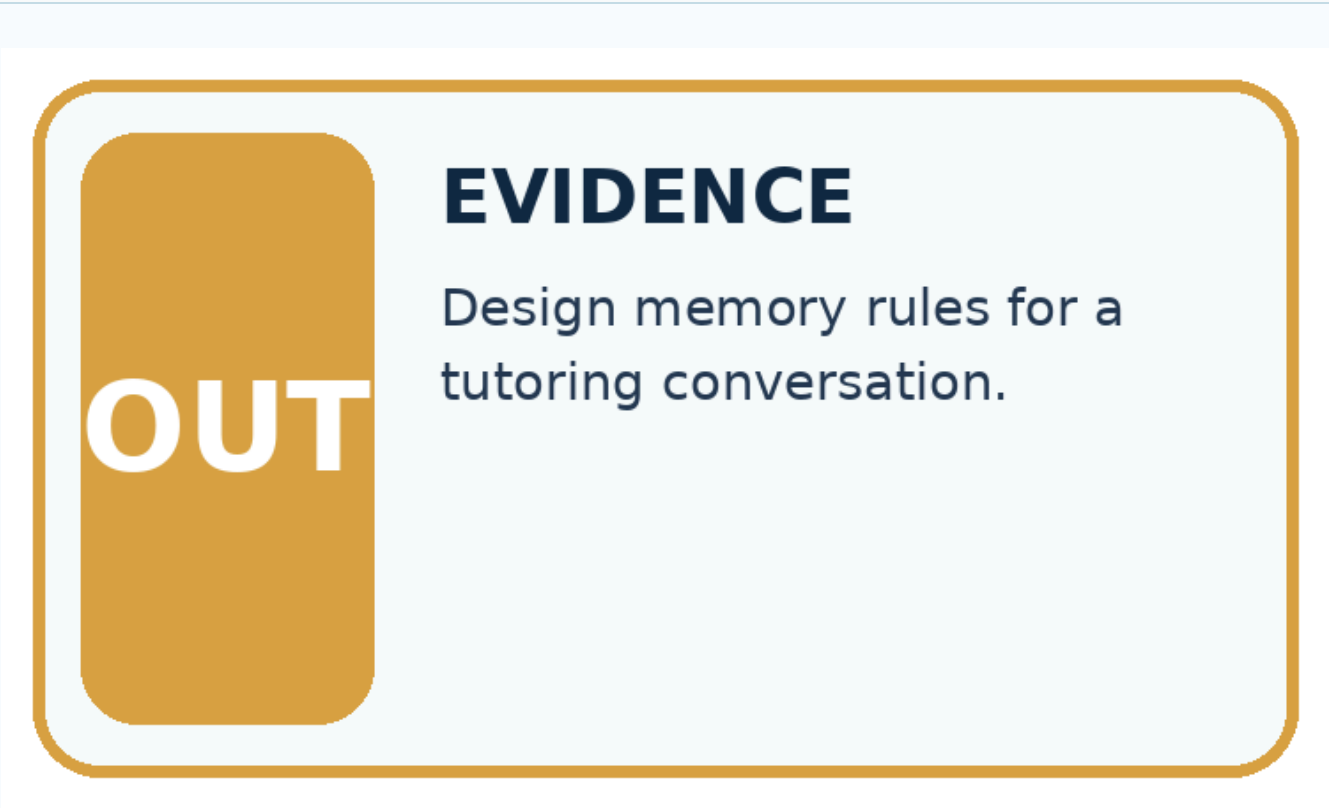
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Design memory rules for a tutoring conversation.

↓


SYSTEM
Products choose what context to keep, summarize, forget, or retrieve.

↓




Which stage could fail even when the other two are correct? Explain:

**Write or draw your reasoning and evidence.**

### A closer explanation

#### Conversation state and retention — evidence trace

| INPUT AND SCOPE | CHECK OR CALCULATION | CONCLUSION AND LIMIT |
| --- | --- | --- |
| Turn 1 selects room A; turn 2 corrects to room B. A later answer uses A from an older summary. | This classroom assistant keeps an approved room ID only for the current session. A later correction replaces the active value but retains an audit event until session end. Do not persist private text merely because it appeared in conversation. | Active state should be B under the correction rule. The summary is stale; regenerate or invalidate it and preserve the correction audit within the declared session lifetime. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

### Connect the picture and the explanation

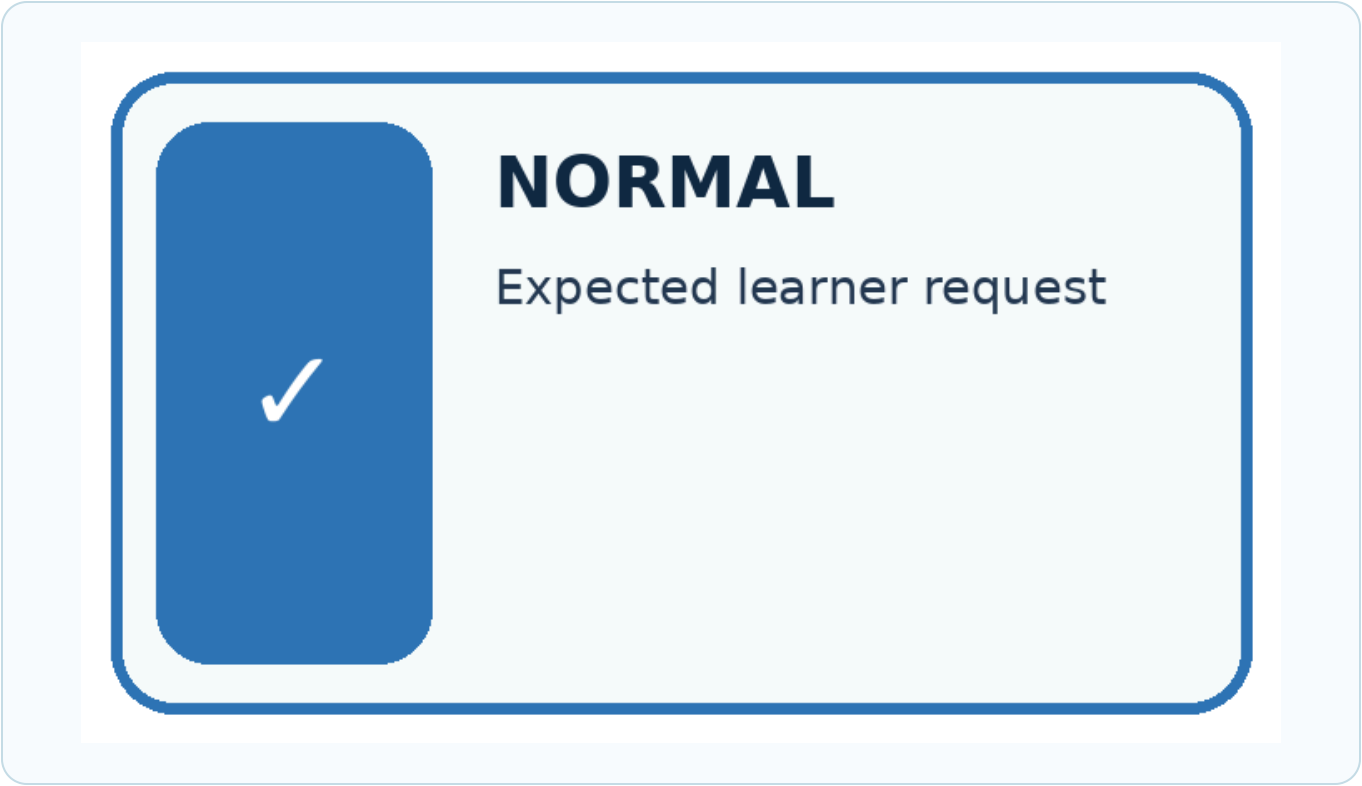
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MAIN LESSON · SECTION 4

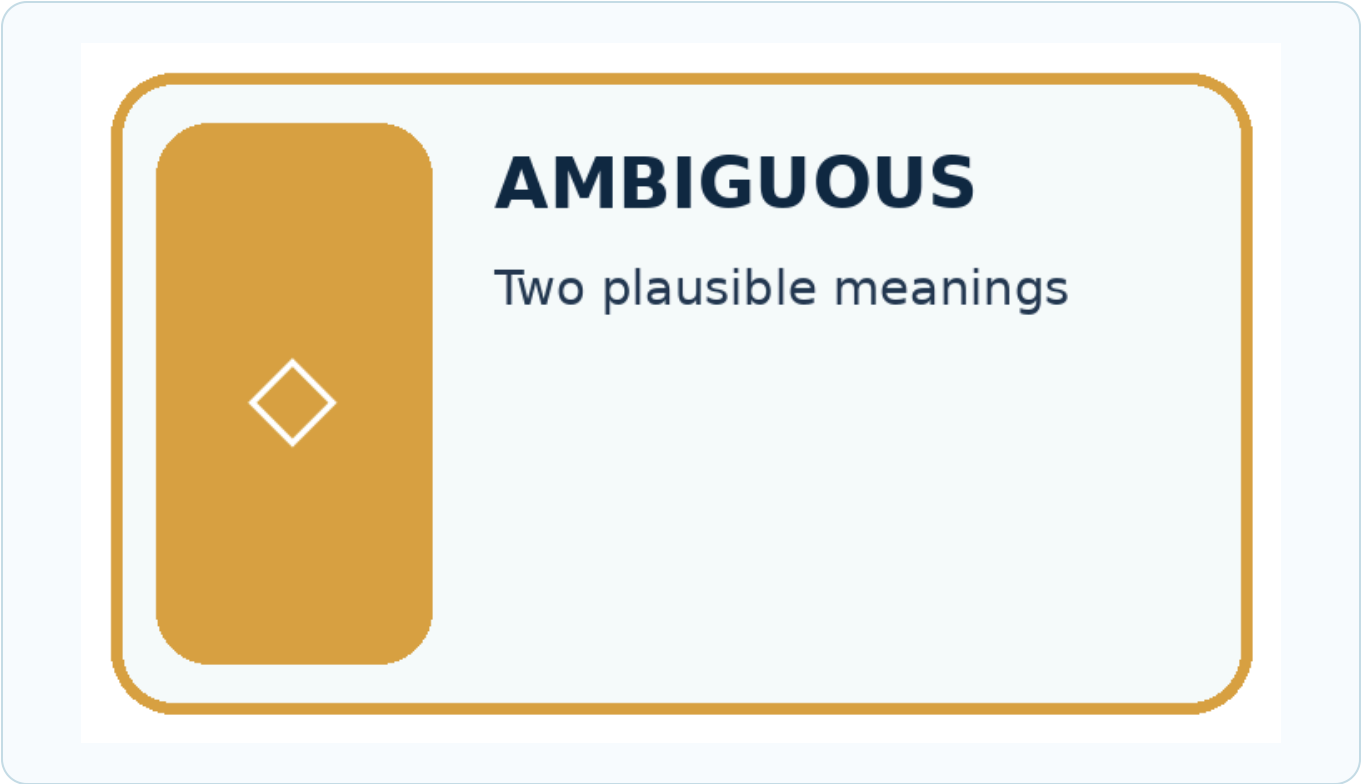
## Compare a current selection with a stale summary

> Two historical choices are not two equally active values.

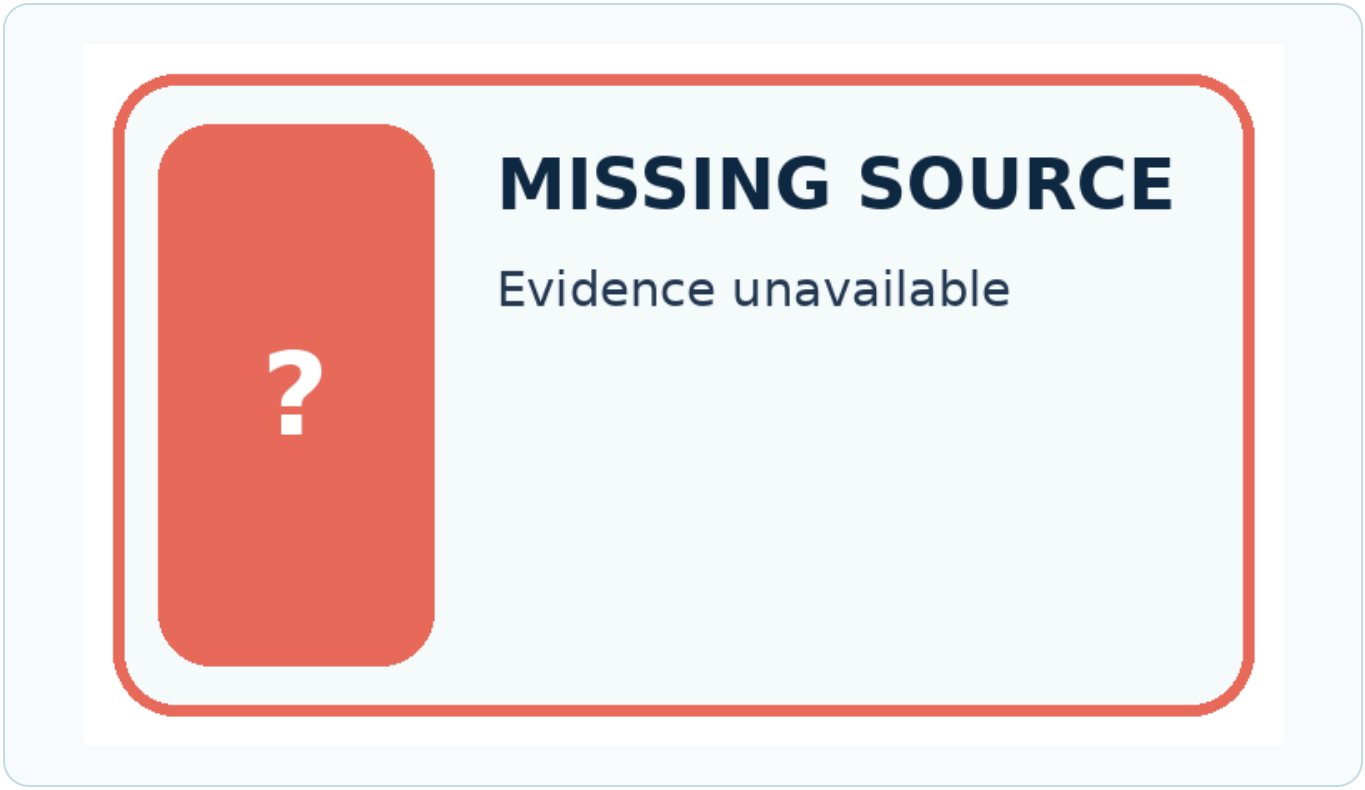
## Four pictures with a purpose



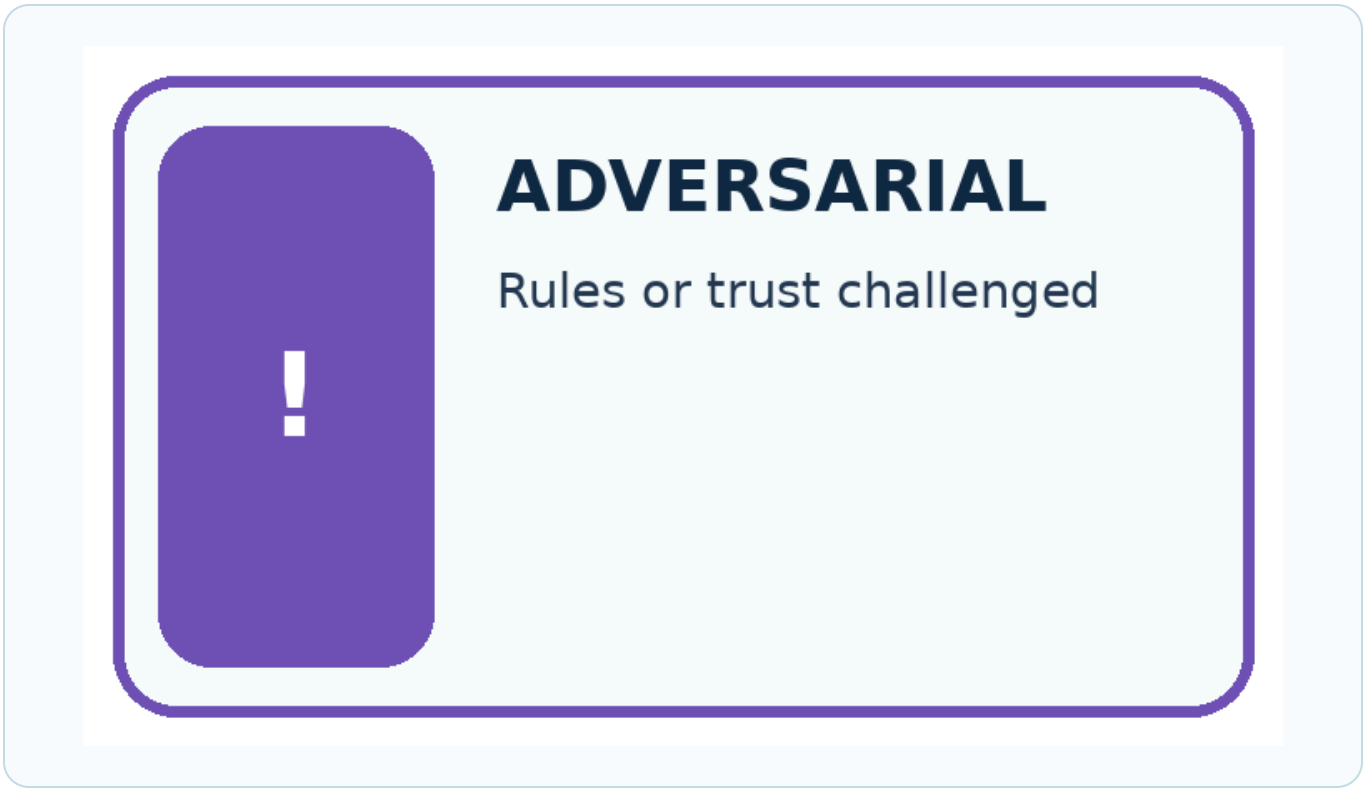

Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.




Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

## Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

## Predict the supplied changed case

- Change: The next session asks “summarise my book” without naming one.

### ▼ Check after predicting

**Prediction**: Ask which book the learner wants.

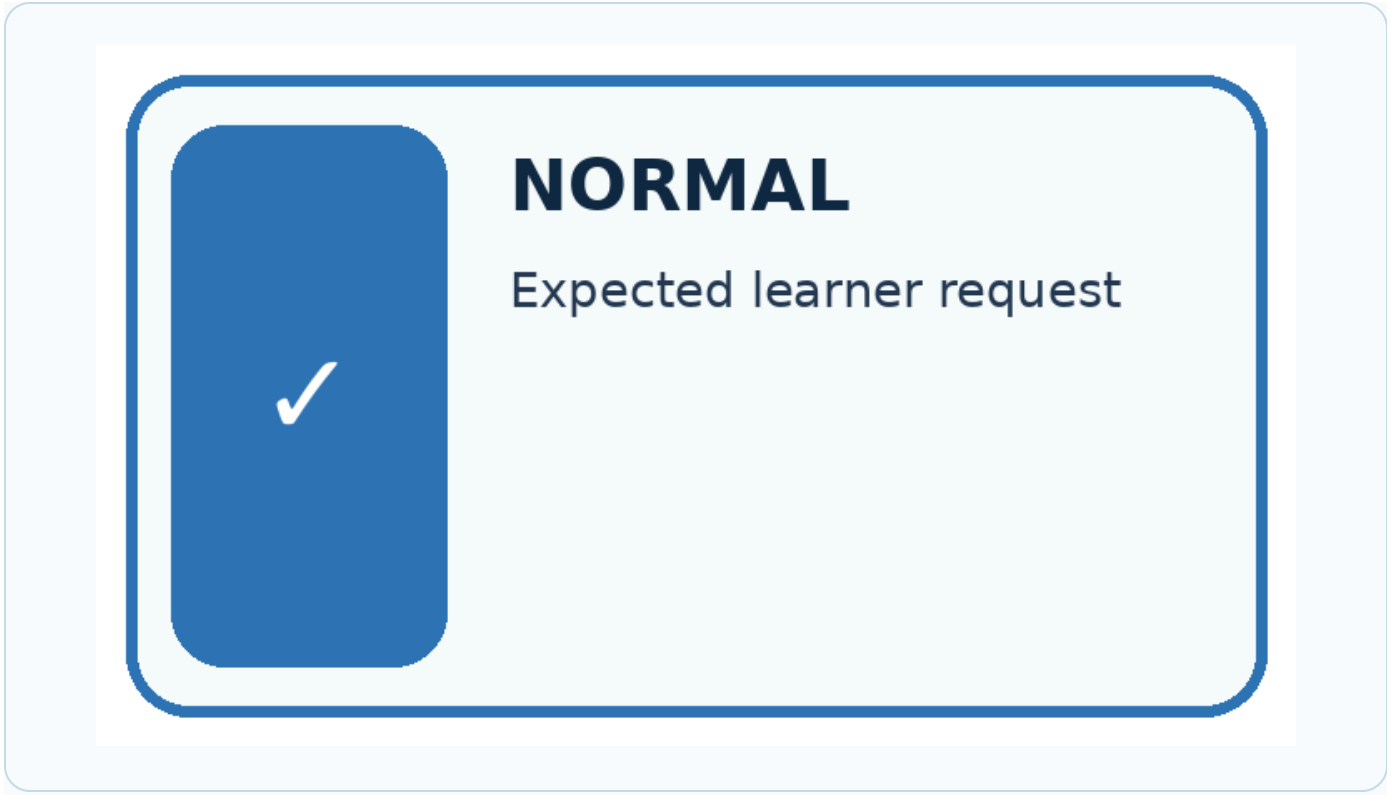
Reason: The declared policy clears **session state**, so yesterday’s choice is unavailable.

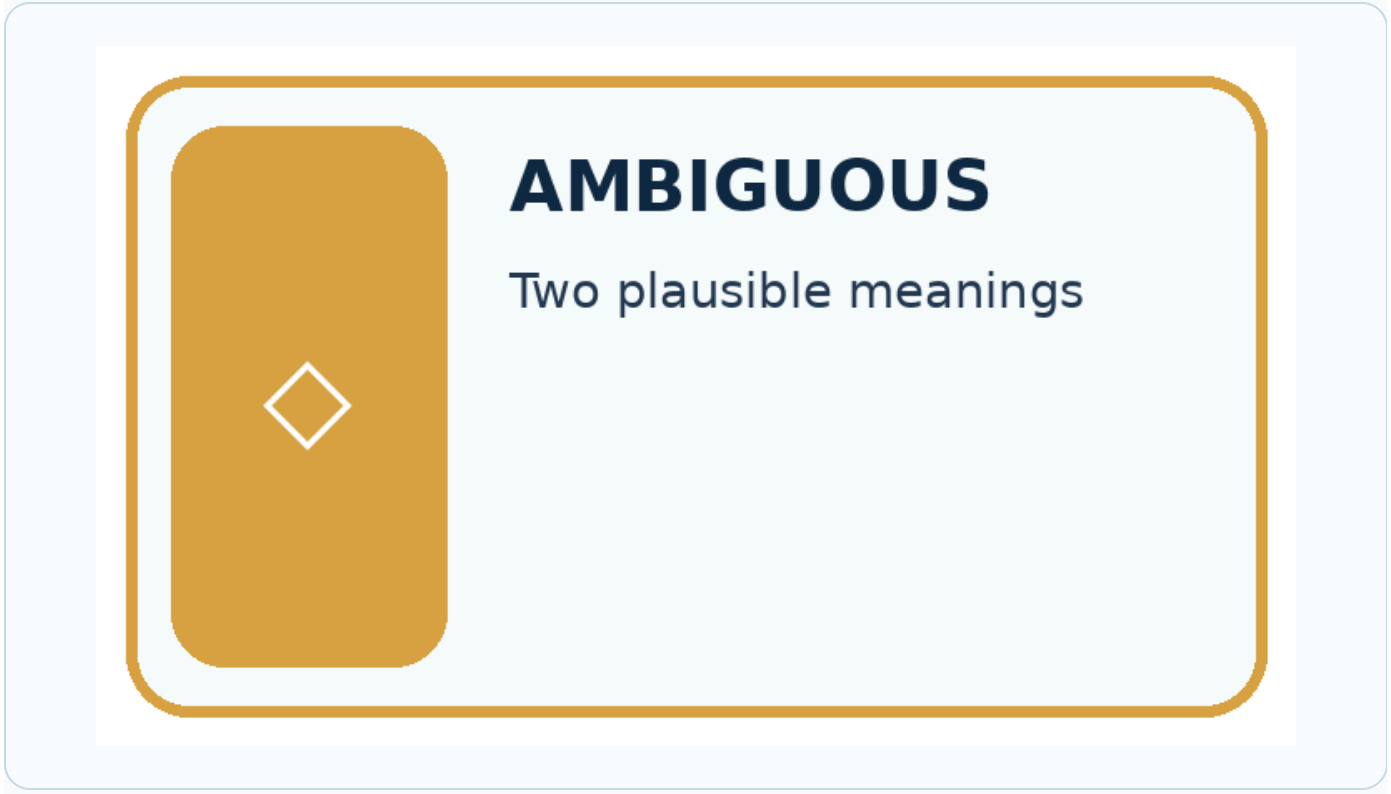
### ▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.




NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.







MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

**Change one thing in the pictured case**

MAIN LESSON · SECTION 6

# Trace Cedar → Maple → no active book

> The new session does not inherit a selection under this policy.

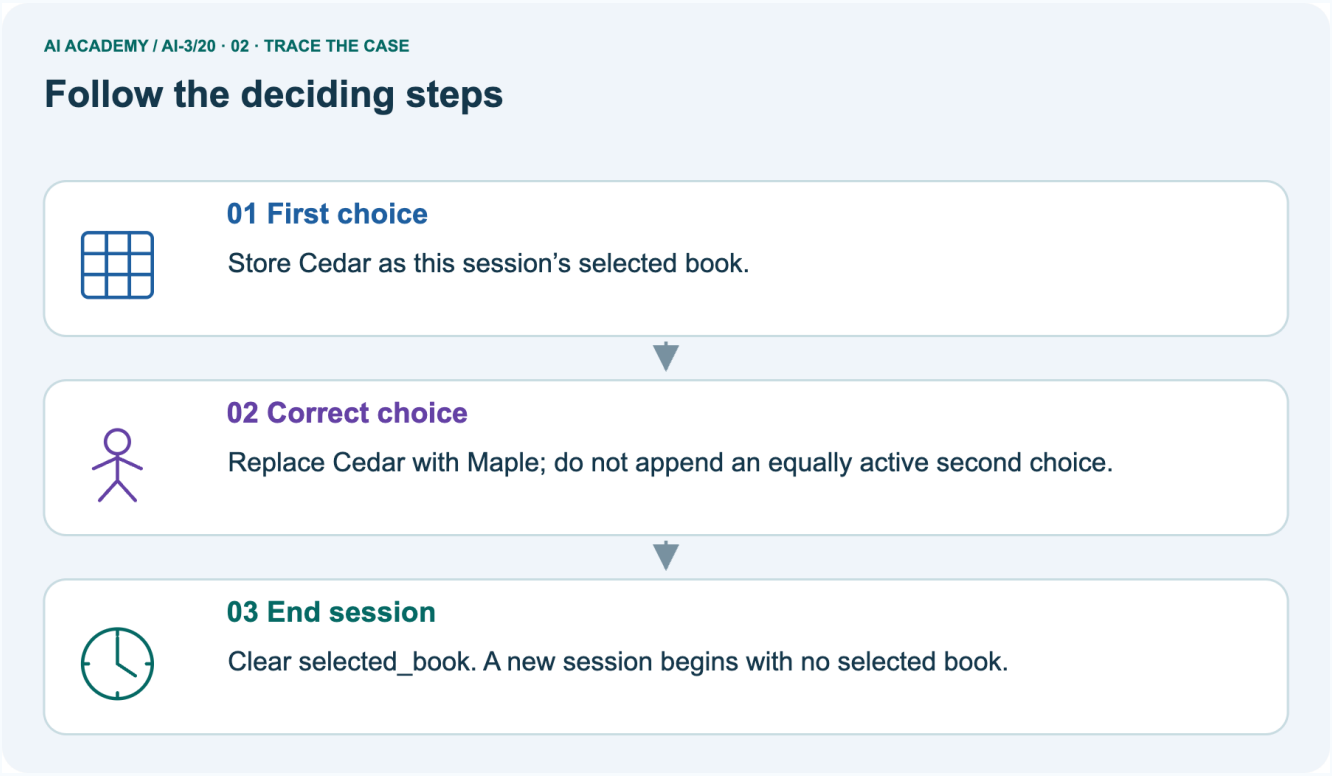
## Read the complete book case

Keep a reading conversation consistent when the learner changes the selected book.

In this toy session, “choose Cedar” sets selected_book=Cedar; “change to Maple” replaces it. **Session state** is cleared on exit.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

## Trace the pictured transitions




Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. First choice | Store Cedar as this session's selected book. |
| 2. Correct choice | Replace Cedar with Maple; do not append an equally active second choice. |
| 3. End session | Clear selected_book. A new session begins with no selected book. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Clear selected_book. A new session begins with no selected book. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

## Inspect each state and lifetime

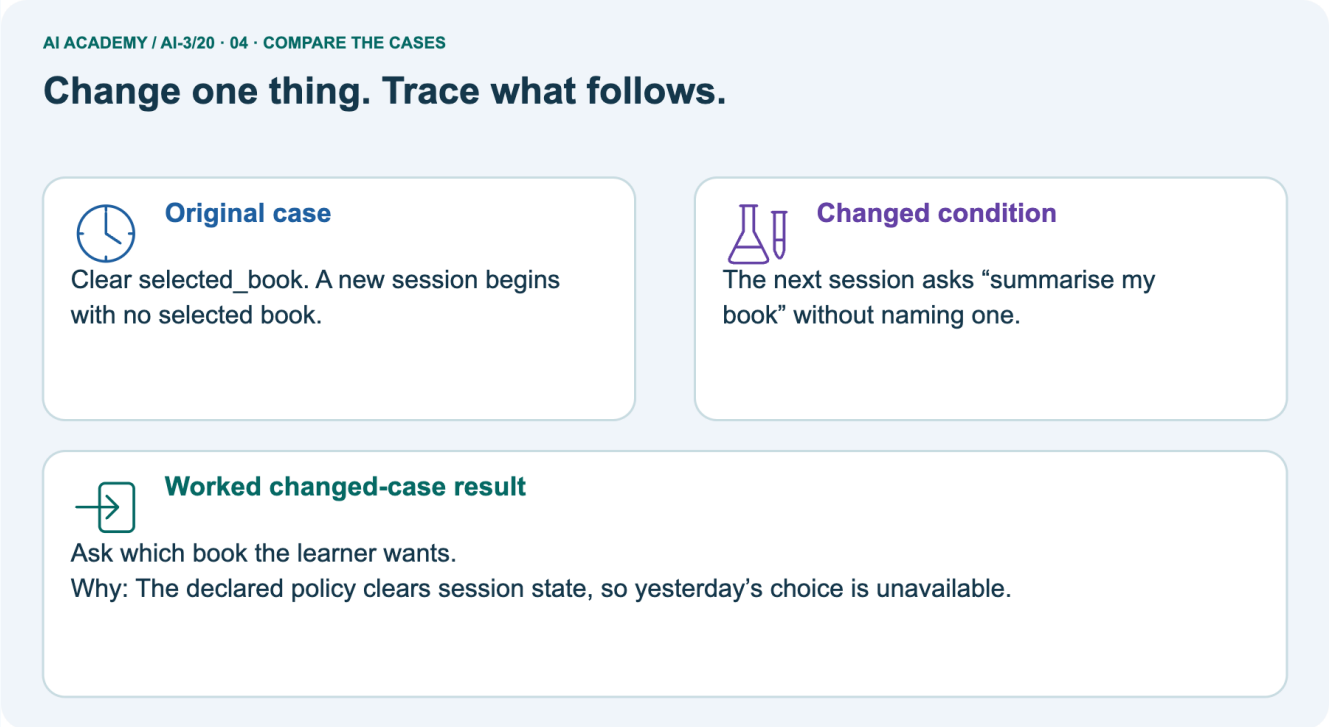
| Event | Active selected_book | Summary / audit action |
| --- | --- | --- |
| Start session | None | No selection to reuse |
| Turn1 choose Cedar | Cedar | Record source turn1 |
| Turn2 change to Maple | Maple | Refresh/invalidate Cedar summary; minimal Cedar→Maple audit |

| Event | Active selected_book | Summary / audit action |
|---|---|---|
| Exit | None | Clear session summary and audit |
| New session asks summarise my book | None | Ask which book; do not silently restore Maple |

## Predict the next-session question

- Change one thing: The next session asks “summarise my book” without naming one. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting




Follow the labelled picture with the matching explanation. Open the vector diagram

## State the application-memory limit

- Conversation **state** is an application record; it is not **evidence** that the model permanently learned the learner’s preferences.

- Read the labelled picture: What is selected_book after the correction and after exit? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

### ▼ Separate practice · Room A corrected to B

Turn 1 selects room A; turn 2 corrects to room B. A later answer uses A from an older summary.

Active **state** should be B under the correction rule. The summary is stale; regenerate or invalidate it and preserve the correction audit within the declared session lifetime.

### ▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

**WEEK 20 • WORKED EVIDENCE**

1. Keep current topic and learner’s chosen difficulty for the session.
2. Summarize progress without storing full private conversation.
3. Forget sensitive or unnecessary details.
4. Retrieve approved study notes only when the learner requests that topic.

1. Keep current topic and learner’s chosen difficulty for the session.

2. Summarize progress without storing full private conversation.

3. Forget sensitive or unnecessary details.

4. Retrieve approved study notes only when the learner requests that topic.

Recalculate, retrace, or re**state** the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

### A closer explanation

Compare the intermediate **evidence** with the final **state**ment. Explain why each number or decision follows from the **state**d **input**s.

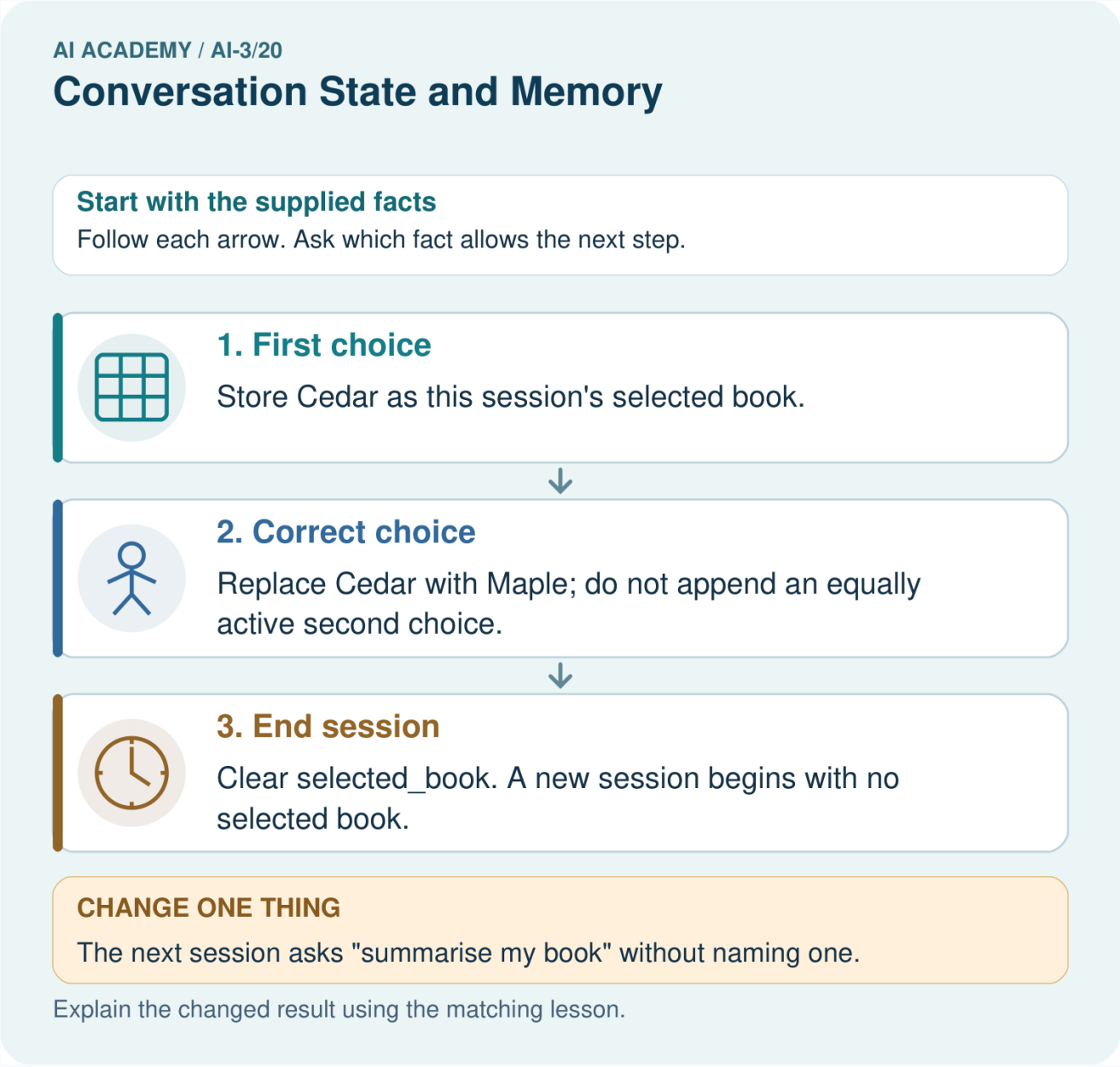
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

## A pictorial worked case: Conversation State and Memory

### The facts we will use




Follow the labelled picture with the matching explanation. Open the vector diagram

### Why this result follows

### What this example does not establish

MAIN LESSON · SECTION 7

# Find the stale value used as authority

> A correction must reach every derived field that the answer uses.

### Repair the stale-summary shortcut

Claim to repair: The summary still says Cedar, so Cedar is the active choice. A derived summary cannot override the later explicit correction to Maple. Refresh it or mark it invalid. Keeping both choices active is also wrong under this one-selection contract. After exit, neither is active; asking for a book is correct missing-state behavior, not proof that the model forgot training data.

## Design a revealing follow-up

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

## COMPARE • DIAGNOSE

Weak idea: Remembering everything always creates a better assistant. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

| Evidence record | Expected | Actual | Likely cause / next test |
|---|---|---|---|
| normal case | works | | |
| boundary case | stated rule | | |
| missing input | safe response | | |
| fresh transfer | generalizes | | |

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

### A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

MAIN LESSON · SECTION 8

# Practise the room correction with less help

> Keep the A/B practice distinct from the fresh L1/L2 assessment.

## The retention and verification limit

Remembered **state**ments are not automatically current or verified; retention should follow the **state**d purpose and policy.

## Complete the room-state W2

### Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Conversation **state** and retention”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

| Plan | Do | Check |
| --- | --- | --- |
| What is given? Which rule or procedure applies? | Record each intermediate step. | Does the result follow? What remains unknown? |

## Audit a generalisation shortcut

- Use the worked case for Conversation **state** and retention. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

### Check your reasoning

## Explain the pictured state transitions

- What is selected_book after the correction and after exit? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

### Explain the picture

**My explanation using a pictured step and the given facts**

## Keep independent work separate

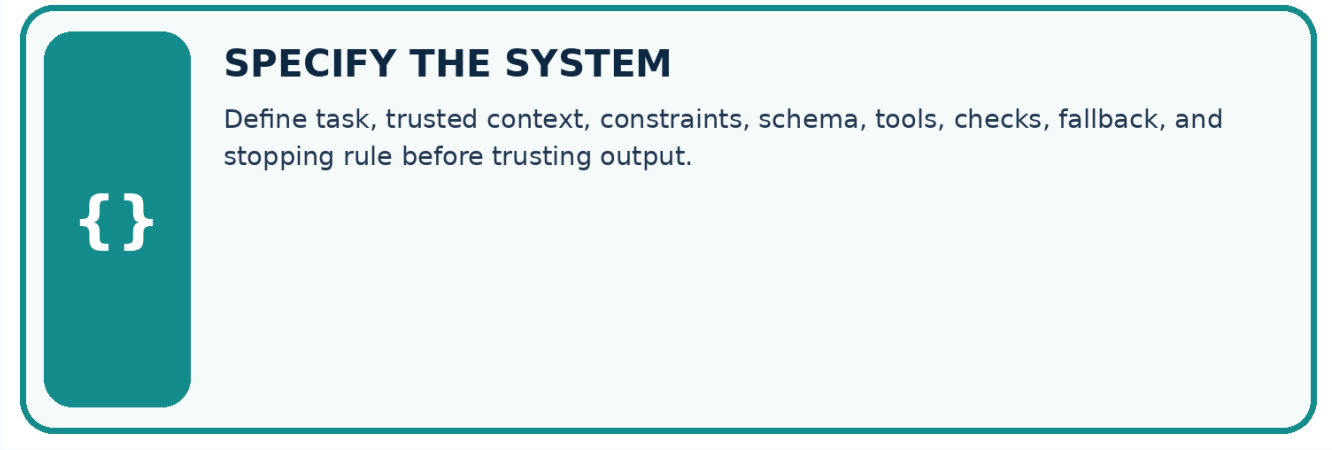
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.




SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the **state** or chooses a path?

> **Write or draw your reasoning and evidence.**

### A closer explanation

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

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MAIN LESSON · SECTION 9

## Sort six complete state and retention cards

Use purpose and lifetime rather than a keep-everything rule.

### All supplied cards and policy guidance

| Supplied fictional information card | Handling under the toy policy |
|---|---|
| A · Current session chooses Cedar at turn1. | Keep selected_book=Cedar with turn provenance until correction or exit. |
| B · Same session corrects book to Maple at turn2. | Replace active selection; minimal audit until exit. |
| C · Cached summary still says Cedar. | Invalidate or regenerate; never treat stale summary as current authority. |
| D · Unrelated private home-address text appears. | Do not copy into task state or summary; no private address is supplied here. |
| E · User asks this Wednesday's library close; no current notice in state. | Retrieve authorised current notice or report unavailable; state is not an approved schedule source. |

| Supplied fictional information card | Handling under the toy policy |
| --- | --- |
| F · Session exits, then new session asks about my book. | Clear session fields and audit; ask which book. |

## Predict, sort and justify the state changes

1. Cover the handling column and sort each complete card into keep, replace, refresh/invalidate, do not retain, retrieve, or clear/ask. Actions can combine; a correction changes active state and a derived summary. Record purpose, provenance and lifetime for every kept field. Then uncover guidance and explain your first mismatch. Retain only a minimal task summary, never the unrelated private text.

   Card E may require retrieval: if the authorised notice is absent, do not invent a time. This is a paper simulation of application state, not a deletion test of a real product. Keep the lab-selection independent case withheld.

## Keep the paper-trace evidence

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

**Test / input:** missing

**Prediction:** _____

**Actual:** _____

**Evidence / calculation:** _____

**Change or keep?:** _____

**Test / input:** fresh transfer

**Prediction:** _____

**Actual:** _____

**Evidence / calculation:** _____

**Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

**Write or draw your reasoning and evidence.**

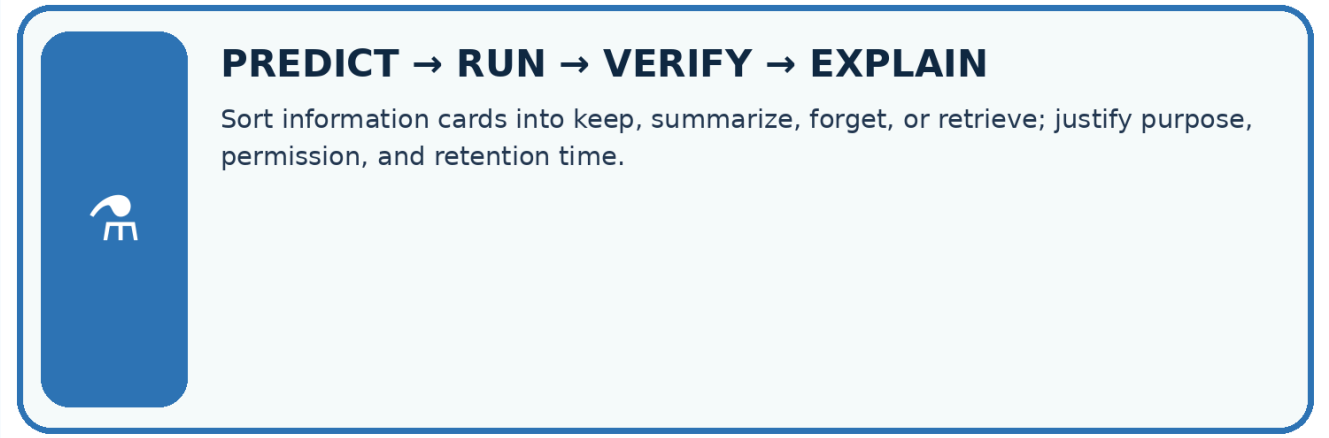
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

**EVIDENCE** LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.




Use the six supplied fictional information cards and explicit session policy. Decide what to keep, replace, refresh, avoid retaining, retrieve or clear; justify purpose, provenance and lifetime. Trace application state without claiming real-service deletion or persistent learning.

**Plan → try → check**

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MAIN LESSON · SECTION 14

## Try alone: trace the fresh lab-selection session

> Apply the declared lifetime, rather than assuming permanent memory.

### Independent events and policy

A session stores lab L1 then corrects to L2; session ends. A new session has no room **input**. **State** the active values at each stage.

Apply the same explicit session-only policy to the supplied lab selection: one active value, correction replaces it, exit clears active state and session audit, and a new session without input has no selection. Trace the stages from the given events; no persistent store or real service is supplied.

## Your first attempt and help record

**My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

## Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

### Independent case: Apply a new case independently

### Week 20 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A learner corrects their project topic from oceans to forests. The next reply still uses oceans. What **state** should change, and what should not be saved unnecessarily?

My answer, with a trace, calculation or supporting **evidence**

**Write or draw your reasoning and evidence.**

### Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

### Connect to your project

Store minimal **session state** and make **persistent memory** optional, visible, and erasable.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

### Transfer practice from the original programme

The conversation **state** is topic=plants, level=Grade 8. The user says 'Keep the topic, explain for Grade 6'. Write the updated **state** and a reset behaviour.

> **Explain the mechanism, evidence and one limit.**

### Use feedback to improve

MAIN LESSON · SECTION 11

# Keep the project's memory policy visible

> Every retained field needs a task purpose and lifetime.

## This week's contribution

For the Verified Study Buddy, list necessary session fields and their sources, correction rule and expiry. Show the current value to the learner, invalidate derived summaries after corrections and provide reset behavior. If designing optional persistent memory later, specify separate visibility, permission and deletion behavior instead of silently extending this session-only policy.

A user's earlier claim still needs verification before becoming a factual answer.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

## Retain the checkpoint and human choice

THIS WEEK'S CHECKPOINT Store minimal **session state** and make **persistent memory** optional, visible, and erasable.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

**Write or draw your reasoning and evidence.**

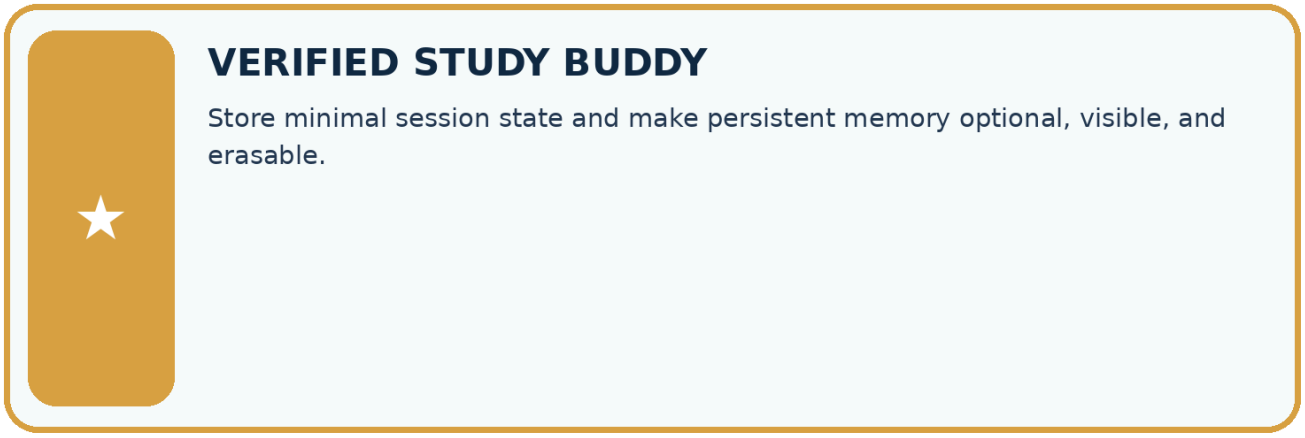
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse fluent **output** with verified truth.

## ▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.




### Evidence for your project

# Finish: state the current value and its lifetime

> A summary, an active field and a learned parameter are different things.

## Explain without notes

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of conversation **state** and retention

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

## Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

## Your quick word guide

> Use the words to explain a particular step, record or limit.

### Core vocabulary

| Word | My own example | Not this / boundary |
|---|---|---|
| state | The current values and information that determine what a process does next. | |
| session memory | Information maintained during the current interaction, without assuming it persists afterwards. | |
| persistent memory | Information retained beyond the current session by an explicit storage mechanism. | |
| retention rule | A policy specifying what information is kept, for how long and when it is deleted. | |

### Terms for the connected explanation

| Word | Meaning |
|---|---|
| Session state | Information retained within a defined interaction lifetime |

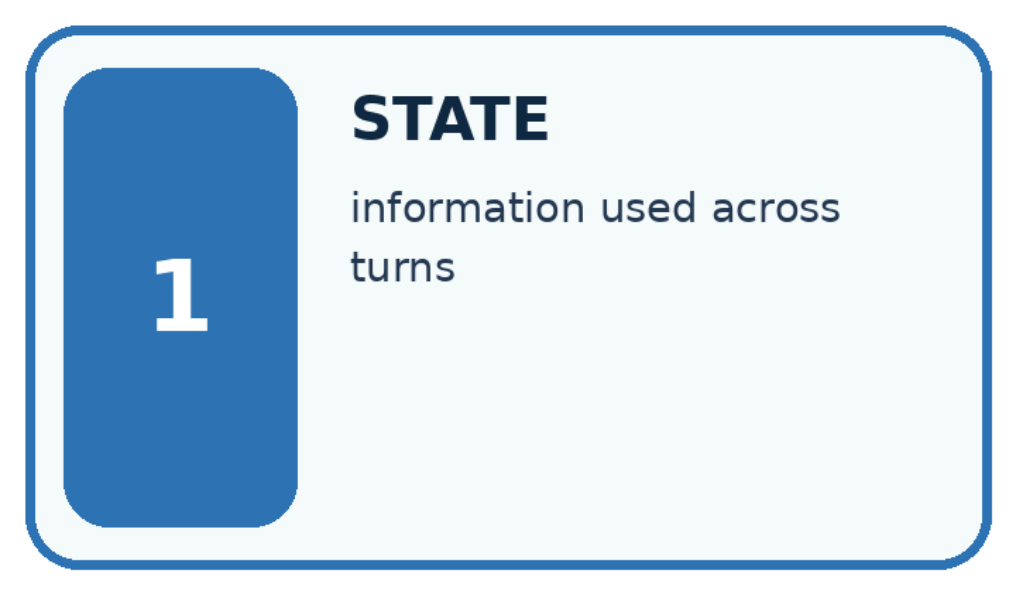
| Word | Meaning |
| --- | --- |
| Correction event | A record updating an earlier active value |

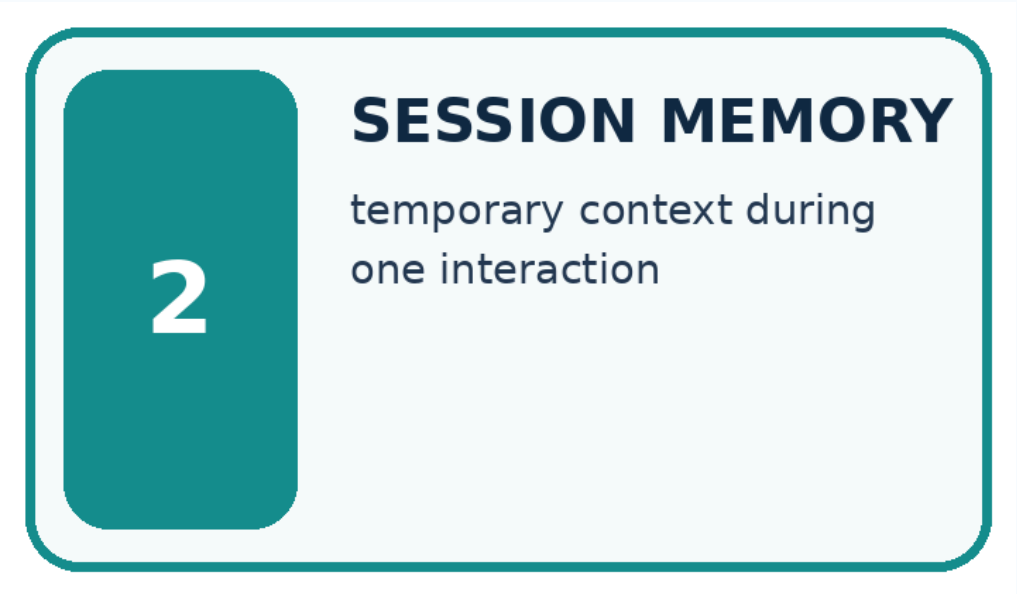
▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
SESSION MEMORY
temporary context during one interaction


3
PERSISTENT MEMOR
stored information across sessions




**Words used in the closer explanation**

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

## Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

### Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, **state**, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

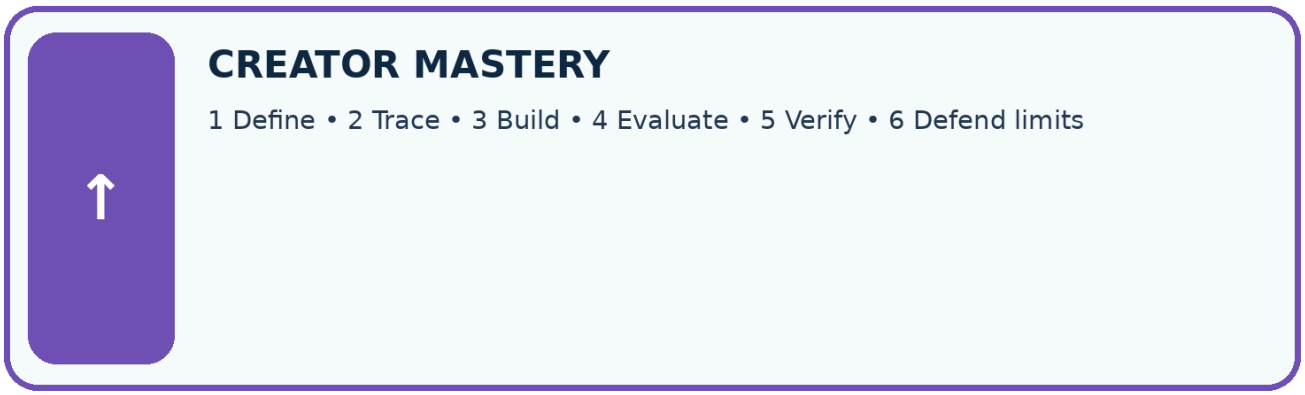
**Write or draw your reasoning and evidence.**

## ▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.




## ▼ Reference: check your explanation

# Reference: check your explanation

> Use after practice. Trace the new lab-selection session independently is the selected independent case.

## Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

## Self-check and record

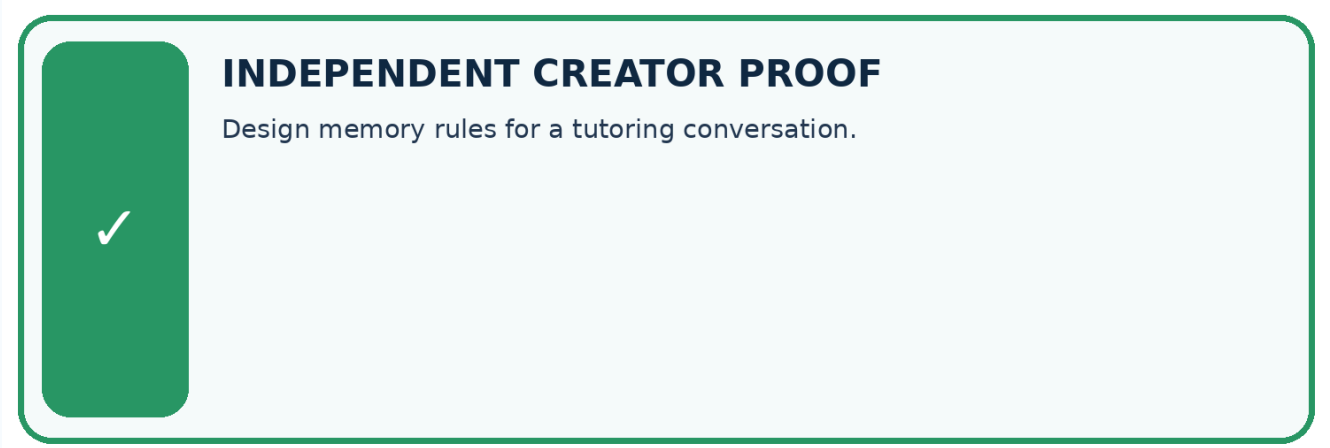
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Trace the new lab-selection session independently is the selected independent case.




EXPLANATION CHECK Require one active value, correction propagation, cleared state after exit and an evidence-seeking new-session response.

**Write or draw your reasoning and evidence.**

▼ **Extra practice: Update only the state the user changed**

REFERENCE · SECTION 13

## Extra practice: Update only the state the user changed

Try the question before opening its worked explanation.

### Read the supplied case and question

**State** stores topic=plants and reading_level=beginner. User requests advanced explanation.

What changes?

## Your explanation

- Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

### ▼ Check after trying

Update reading level, retain topic unless changed. Keep only necessary **state** with correction/reset. Context does not guarantee permanent memory.

| Evidence or stage | Value or result |
|---|---|
| Before | topic=plants, level=beginner |
| Request | Use an advanced explanation |
| After | topic=plants, level=advanced |
| Reset | Clear stored fields when requested |

▼ Earlier wording, alternatives and picture transcripts

**Week 20 Worked case**

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

**The situation**

**The question**

**Worked explanation**

**Explain it in your own words**

**Trace the supplied case visually**

**Update only the state the user changed**

**State** stores topic=plants and reading_level=beginner. User requests advanced explanation.

What changes?

Update reading level, retain topic unless changed. Keep only necessary **state** with correction/reset. Context does not guarantee permanent memory.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

## Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

## Targets and evidence

| By the end, I can | How I will show it |
| --- | --- |
| Explain the mechanism and terminology of conversation state and retention | Explain the deciding step, show a trace or test, and state one limit. |
| Apply the stated procedure to a new case with a traceable result | Explain the deciding step, show a trace or test, and state one limit. |
| Identify an assumption or limitation and design a revealing follow-up | Explain the deciding step, show a trace or test, and state one limit. |

## Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

### ▼ Earlier wording, alternatives and picture transcripts

Start with this question: Keep a reading conversation consistent when the learner changes the selected book.

### ▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

# Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

## First supported question

- Keep a reading conversation consistent when the learner changes the selected book.
- Which labelled step matches this detail? Clear selected_book. A new session begins with no selected book.
- 1. First choice
- 2. Correct choice
- 3. End session
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.
- After your attempt, compare with the supported explanation: End session Clear selected_book. A new session begins with no selected book. Read the picture in order. Clear selected_book.

  A new session begins with no selected book. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

### Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the first explanation

**Practice 2: explain the deciding evidence**

## Changed-case question

- Changed-case practice: The next session asks “summarise my book” without naming one. Which conclusion is supported by the supplied facts?
- 1. Ask which book the learner wants.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.
- After your attempt, compare with the supported explanation: Ask which book the learner wants. The declared policy clears **session state**, so yesterday’s choice is unavailable.

**My selected conclusion, deciding fact and explanation**

▼ Check the changed case

**Repair a missing building block**

## Repair and save your work

Review User Need and AI Contribution (ai-3/19). Goal: Explain the **mechanism** and terminology of capstone requirements and corpus. Short practice: What non-AI baseline would reveal whether this helper adds value?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Model Memory vs Retrieval (ai-3/17). Goal: Explain the **mechanism** and terminology of retrieval and indexing. Short practice: Why can the assistant not certify 15: 00 when the notice is missing?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Hallucination Failure Types (ai-3/14). Goal: Explain the **mechanism** and terminology of generation failures and verification. Short practice: Can we conclude there will definitely be no lunch from S alone?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

**Keep your completed work**

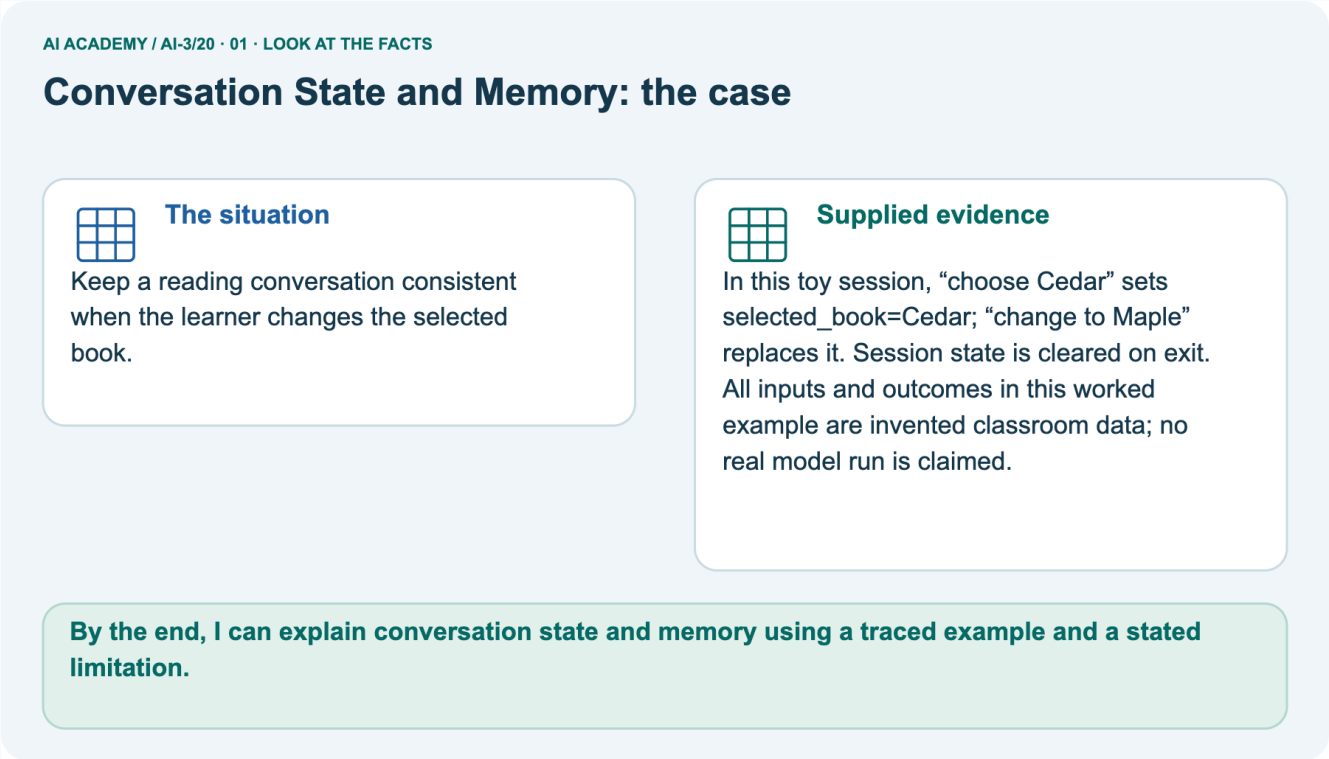
▼ **Reference: the six-picture journey**

REFERENCE · SECTION 18

## Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

## 01. Conversation State and Memory: the case



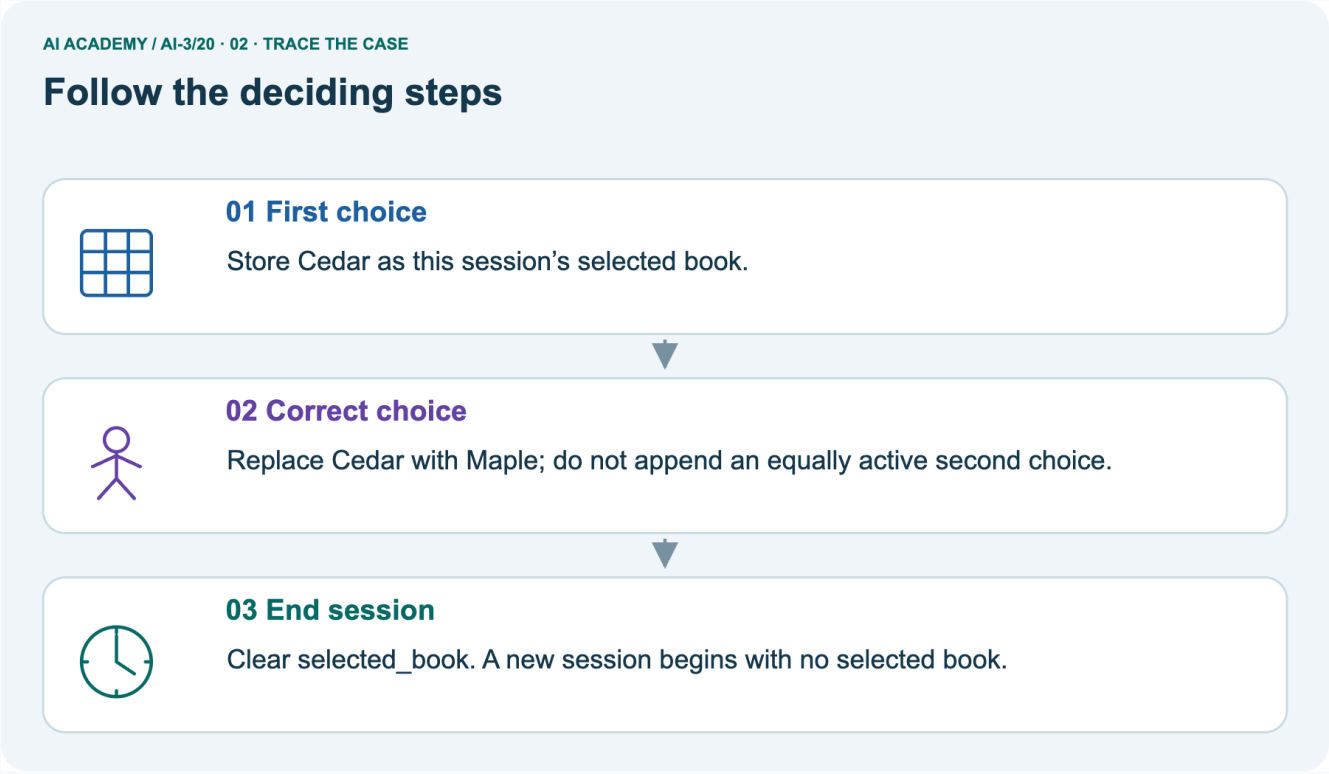

Follow the labelled picture with the matching explanation. Open the vector diagram

Keep a reading conversation consistent when the learner changes the selected book. Supplied facts: In this toy session, “choose Cedar” sets selected_book=Cedar; “change to Maple” replaces it. **Session state** is cleared on exit.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

## 02. Follow the deciding steps



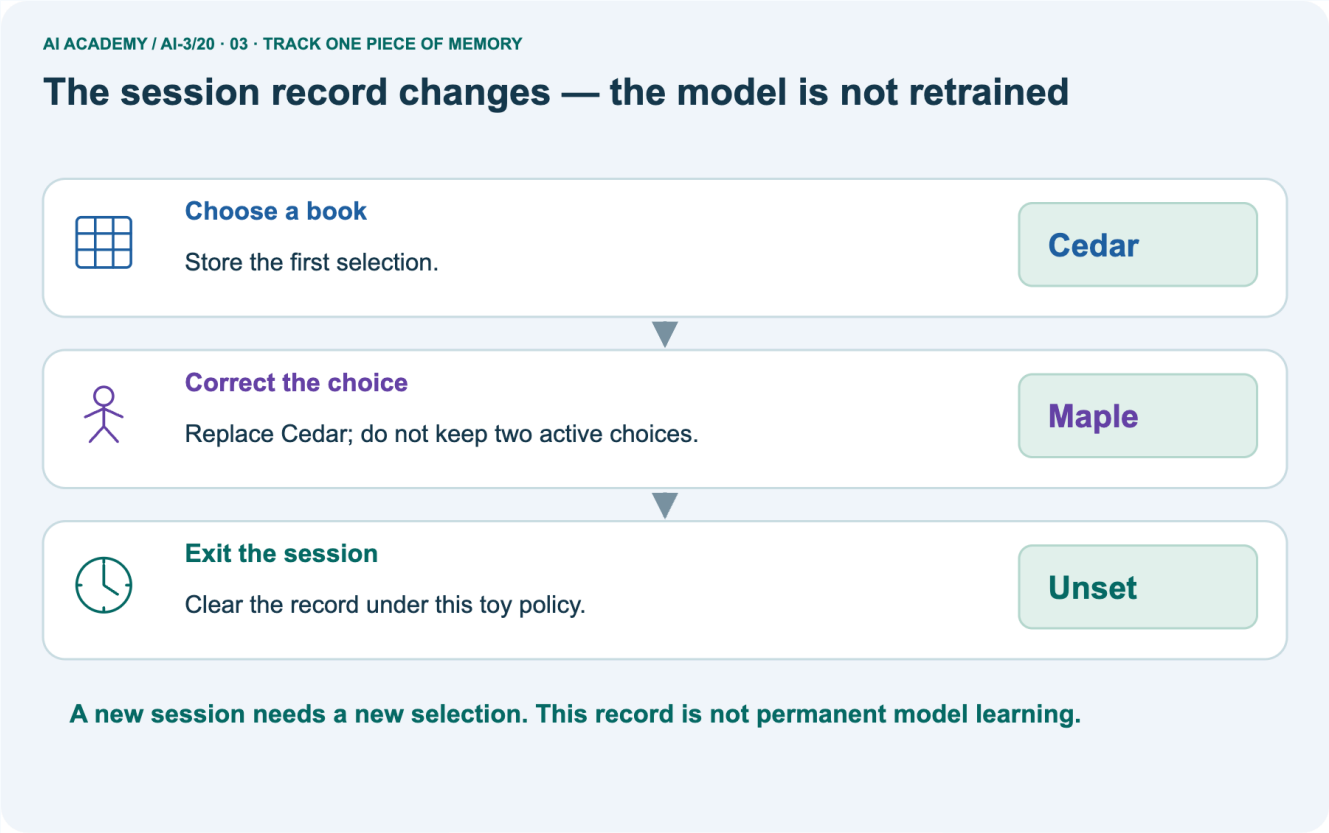

Follow the labelled picture with the matching explanation. Open the vector diagram

First choice: Store Cedar as this session's selected book. → Correct choice: Replace Cedar with Maple; do not append an equally active second choice. → End session: Clear selected_book.

A new session begins with no selected book.

Point to each numbered stage and explain which recorded **input** it uses.

## 03. The session record changes — the model is not retrained

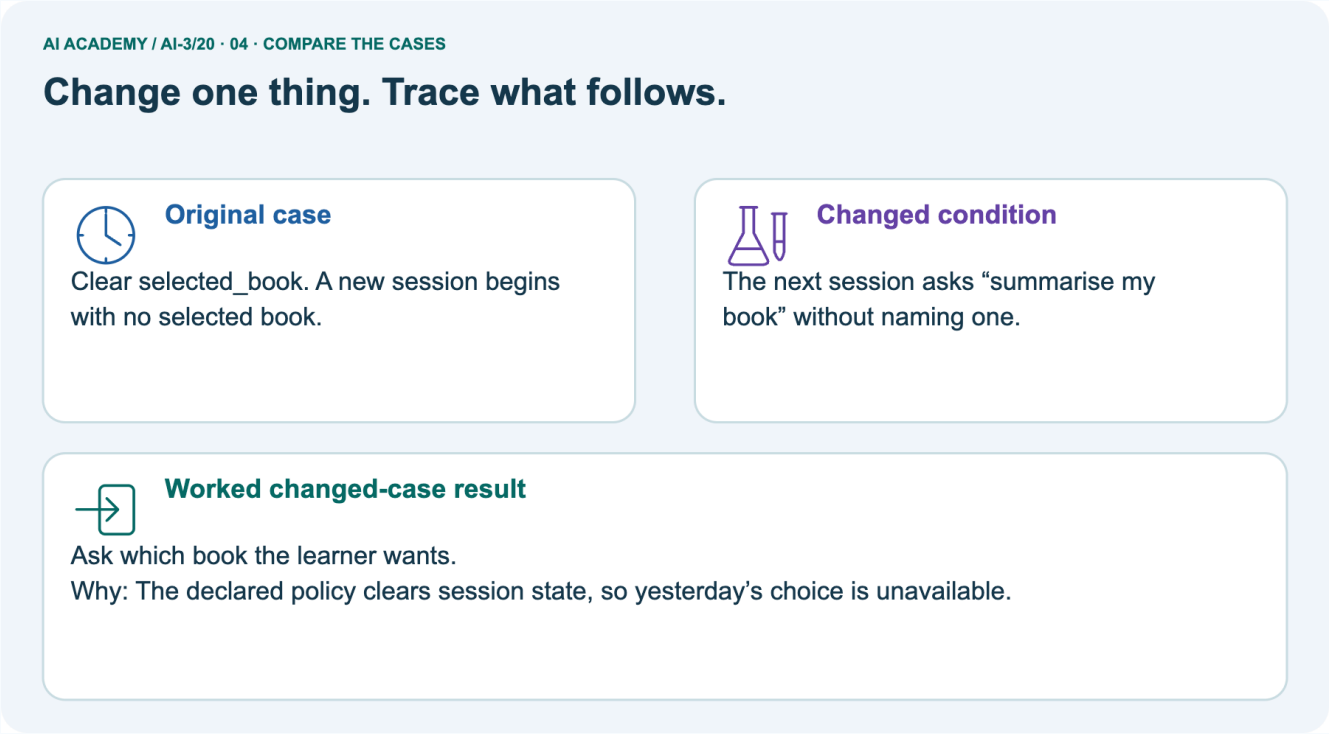



Follow the labelled picture with the matching explanation. Open the vector diagram

The selected_book **session state** is Cedar after the first choice, Maple after correction, and unset after exit. This is an application record, not proof of permanent learning.

Read the labels and values. Explain the relationship the model shows.

## 04. Change one thing. Trace what follows.



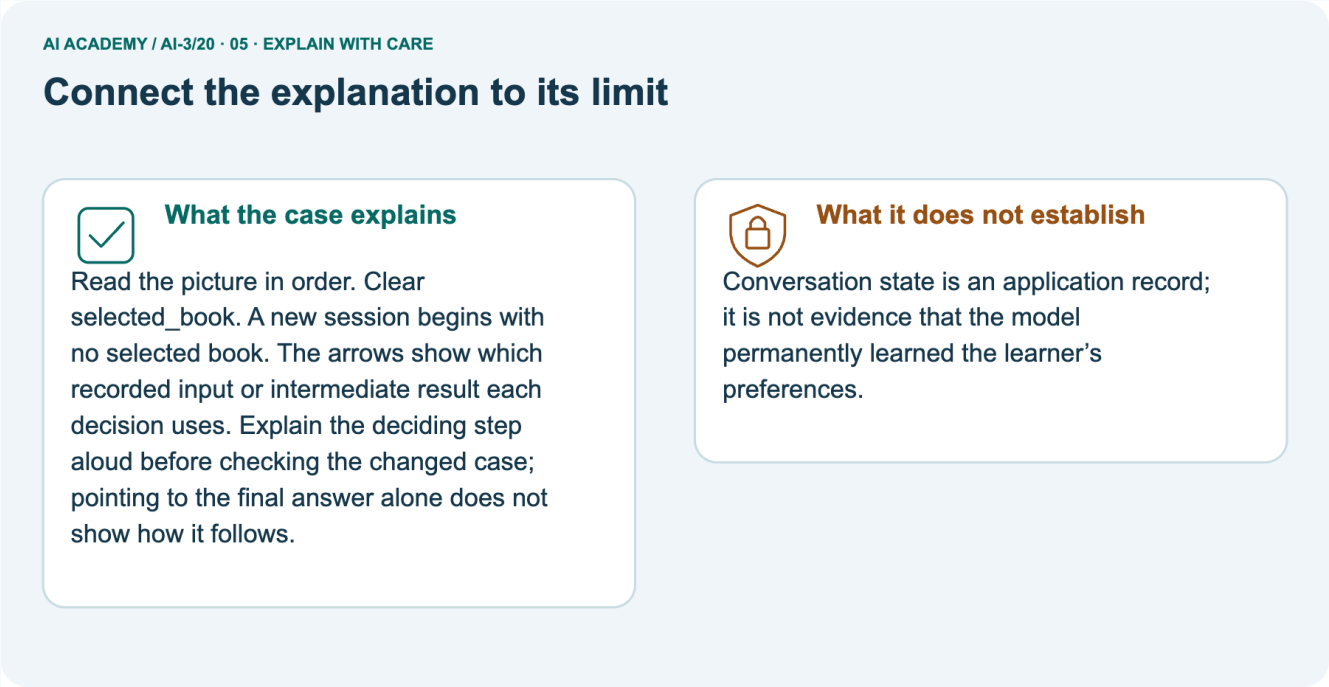

Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Clear selected_book. A new session begins with no selected book. Changed condition: The next session asks “summarise my book” without naming one. Expected result: Ask which book the learner wants.

Reason: The declared policy clears **session state**, so yesterday’s choice is unavailable.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

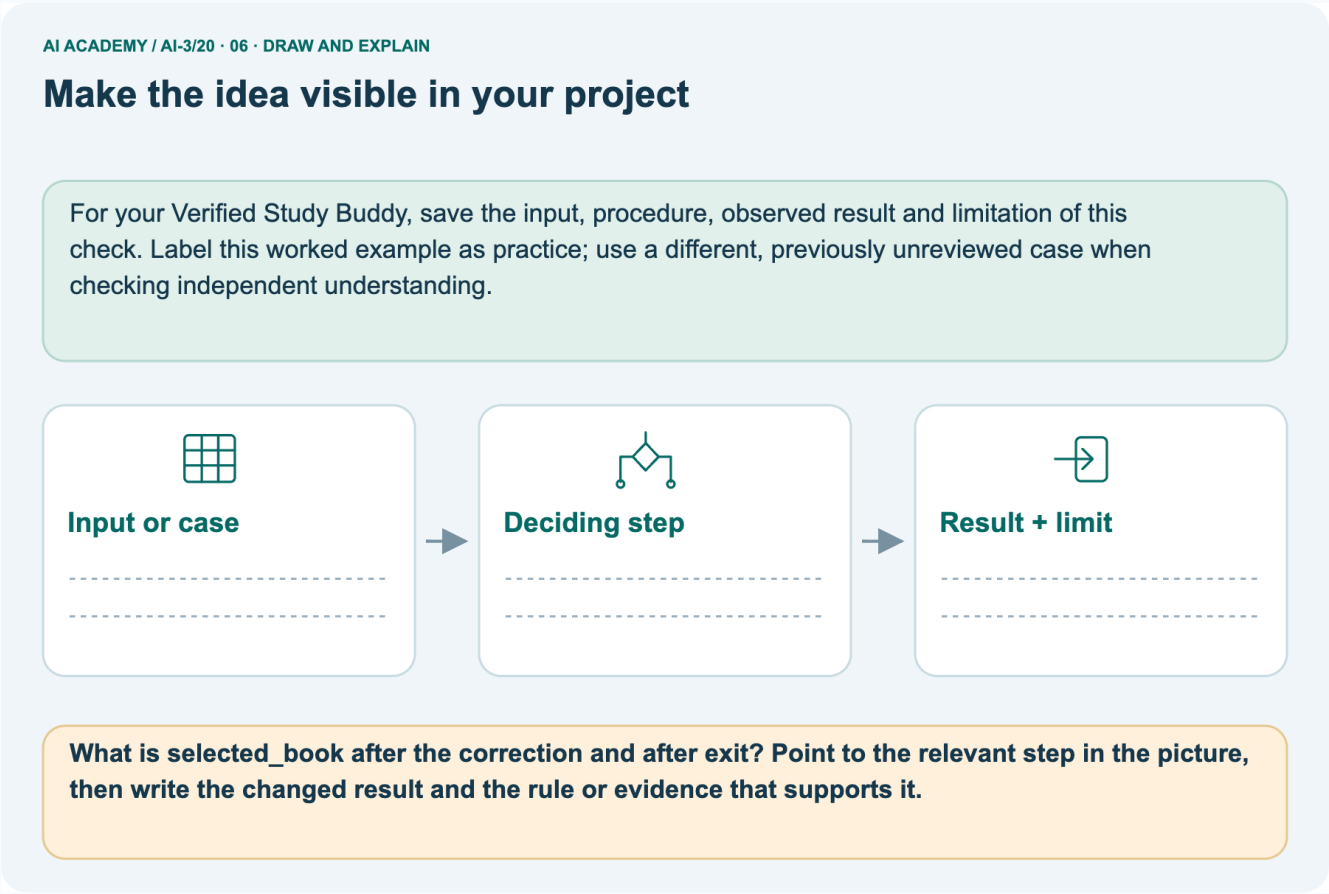
Read the picture in order. Clear selected_book. A new session begins with no selected book. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Conversation **state** is an application record; it is not **evidence** that the model permanently learned the learner's preferences.

Explain the case, then point to a limit on the claim.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What is selected_book after the correction and after exit? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

## ▼ Earlier wording, alternatives and picture transcripts

**01. Conversation State and Memory: the case**

**02. Follow the deciding steps**

**03. The session record changes — the model is not retrained**

**04. Change one thing. Trace what follows.**

**05. Connect the explanation to its limit**

**06. Make the idea visible in your project**

## ▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.